

The permanent spectrum of cubic zero–one matrices of order twelve

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Abstract

The permanents of the 12×12 zero–one matrices having three ones in every row and column take exactly 189 distinct values. We determine the full spectrum by computing perfect-matching counts for connected cubic bipartite graphs and then taking all component products. The connected spectrum has 148 values. Completeness of the graph lists is certified by pairwise nonisomorphism and an orbit-mass identity against an independently computed labeled-matrix count. All 34 623 connected graphs needed at part sizes 3 through 12 were checked by separate exact implementations. The accompanying data contain the complete frequency distribution and a matrix realizing each of the 189 values.

1 Statement and comparison

For $n \geq 3$, let

$$\mathcal{M}_{n,3} = \left\{ A \in \{0,1\}^{n \times n} : \sum_{j=1}^n A_{ij} = 3 \ (1 \leq i \leq n), \quad \sum_{i=1}^n A_{ij} = 3 \ (1 \leq j \leq n) \right\}.$$

Define $S_n = \{\text{per}(A) : A \in \mathcal{M}_{n,3}\}$, where $\text{per}(A) = \sum_{\sigma \in \mathfrak{S}_n} \prod_{i=1}^n A_{i,\sigma(i)}$. The definition permits repeated rows and columns and disconnected bipartite graphs.

Theorem 1. *For the unrestricted family $\mathcal{M}_{12,3}$, one has $|S_{12}| = 189$. The exact set is displayed in Appendix A.*

OEIS A185178 records this cardinality problem [1]. The saved entry, revision 42 dated May 30, 2026, lists the values

$$(|S_3|, \dots, |S_{11}|) = (1, 1, 2, 4, 8, 18, 32, 61, 109)$$

and the bound $|S_{12}| \geq 170$. Its linked examples and frequency tables are due to Robert P. P. McKone [2, 3]. We reproduce every entry of the complete labeled frequency tables for $n = 3, \dots, 11$. At order twelve, the exact determination here contains 19 values absent from the saved 170-matrix list:

$$\begin{aligned} &289, 296, 333, 340, 354, 378, 390, 396, 403, 405, \\ &408, 416, 432, 468, 504, 612, 648, 702, 729. \end{aligned} \tag{1}$$

Only 296 among these occurs in a connected graph of part size twelve. Thus extending the connected list alone does not describe the whole comparison. Our contribution is the exhaustive determination, with a certificate excluding further values. The comparison is specifically to the

primary computational artifacts saved on September 8, 2026; it is not a claim of comprehensive bibliographic priority.

For adjacent context, permanent-rank methods also occur in the work of Zeng, Liu, and Ratnavelu [4]. Exact finite arithmetic combined with combinatorial coefficient arguments appears in Zeng's work on distinct cycle lengths [5]. These references concern different problems and are cited only for this broader context; neither supplies a step of the determination of S_{12} .

2 Components and labeled counts

The bipartite graph of $A \in \mathcal{M}_{n,3}$ has its rows and columns as the two vertex parts, with an edge for each entry equal to one. It is simple and cubic, and its perfect matchings are precisely the nonzero permanent summands. In each component the two part sizes are equal, because counting edges gives three times either size. Simplicity forces their common size to be at least three. A connected bipartite graph has exactly two ordered bipartitions. Consequently uncolored graph isomorphism classes suffice: interchanging the parts transposes the matrix and preserves its permanent.

Let C_k be the perfect-matching spectrum of connected simple cubic bipartite graphs on $2k$ vertices. Products of sets below mean products of their elements. Independence of matchings in distinct components gives

$$S_n = \bigcup_{\substack{k_1 + \dots + k_t = n \\ 3 \leq k_1 \leq \dots \leq k_t}} C_{k_1} \cdots C_{k_t}. \quad (2)$$

Conversely, each product in this union is attained by a block diagonal matrix. Repeated components are allowed. This proves both inclusions in (2).

To certify the coverage of each connected graph list, we count labeled matrices without using that list. Write $T_n = |\mathcal{M}_{n,3}|$, with $T_0 = 1$ and $T_1 = T_2 = 0$. After some initial rows have been filled, use a state (a, b, c, d) recording the numbers of columns of current sums 0, 1, 2, 3, respectively. Initialize the state $(n, 0, 0, 0)$ with weight one. A new row chooses x, y, z columns of current sums 0, 1, 2, where $x + y + z = 3$, and gives the transition

$$(a, b, c, d) \mapsto (a - x, b + x - y, c + y - z, d + z) \quad \text{with weight} \quad \binom{a}{x} \binom{b}{y} \binom{c}{z}. \quad (3)$$

Only nonnegative feasible choices are used. Although states aggregate column labels, the binomial factors count the actual choices among labeled columns. Every partial matrix of a given state has the same number of transitions of each type. Induction over the rows therefore shows that after n transitions the weight of $(0, 0, 0, n)$ is exactly T_n .

Let D_n count matrices with connected graph. For $n \geq 1$, distinguish the component containing the first row. If its part size is k , choose the other rows and its columns, then the connected matrix on these labels and the remaining matrix. This is a bijective construction, giving

$$T_n = \sum_{k=1}^n \binom{n-1}{k-1} \binom{n}{k} D_k T_{n-k}. \quad (4)$$

The coefficient of D_n is one, so subtraction determines D_n successively. The exact totals at the required order are

$$T_{12} = 9\,254\,768\,770\,160\,124\,288\,000, \quad (5)$$

$$D_{12} = 9\,253\,919\,577\,117\,156\,480\,000. \quad (6)$$

A separate coefficient calculation checks the state recurrence. Put $p_j = \sum_{i=1}^n x_i^j$ and let e_3 be the third elementary symmetric polynomial. Then

$$T_n = [x_1^3 \cdots x_n^3] e_3(x_1, \dots, x_n)^n, \quad 6e_3 = p_1^3 - 3p_1p_2 + 2p_3.$$

Expanding by $a + b + c = n$ gives the independent formula

$$T_n = 6^{-n} \sum_{a+b+c=n} \binom{n}{a, b, c} (-3)^b 2^c \frac{n! (3a+b)!}{a! 6^a}. \quad (7)$$

Indeed, the b squared variables from p_2^b and the c cubed variables from p_3^c must be distinct and disjoint. Their weighted choices number $n!/a!$. The remaining exponents in p_1^{3a+b} are three at the a untouched variables and one at the b squared variables, giving $(3a+b)!/6^a$. All divisions in this check are performed exactly. Formula (7) agrees with (3) for every $0 \leq n \leq 12$, using $T_0 = 1$ separately.

3 A certificate of census completeness

Lemma 1. *Let $\mathcal{G}_k$ be a list of pairwise nonisomorphic connected simple cubic bipartite graphs with part size k . If*

$$\sum_{G \in \mathcal{G}_k} \frac{2(k!)^2}{|\text{Aut}(G)|} = D_k, \quad (8)$$

then the list contains every such connected graph up to uncolored isomorphism.

Proof. For a fixed G , there are $2(k!)^2$ choices of an ordered bipartition and bijections of the two parts with the row and column labels. The full uncolored automorphism group acts freely on these choices. Two choices give the same labeled matrix exactly when an automorphism relates them. Thus the number of distinct labeled matrices represented by G is $2(k!)^2/|\text{Aut}(G)|$. This also covers graphs with no automorphism interchanging the parts. For example, $K_{3,3}$ contributes $2(3!)^2/72 = 1$.

Nonisomorphic graphs represent disjoint subsets of the connected labeled matrices. Equation (8) says these disjoint subsets already contain all D_k matrices. An omitted class would contribute a positive additional number, which is impossible. $\square$

For $k = 3, \dots, 11$, the input lists were generated with `geng -b -c -d3 -D3` from `nauty/Traces 2.9.3`. For $k = 12$, the input is the 24-vertex connected cubic bipartite archive from House of Graphs [6], attributed there to `minibaum`. Each record was checked for simplicity, degree three, connectivity, bipartiteness, and equal part sizes. The certification by Lemma 1 does not assume that either generator produced a complete list.

The initial computation used `nauty` canonical forms and automorphism orders, with `Traces` as a second canonicalization and exhaustive integer permutation search as a check of every automorphism order. A further implementation, using neither library nor the initial code, recomputed canonical forms and automorphism orders as follows.

Start with one ordered cell containing all vertices. Refine each cell by the sorted triple of the neighbor-cell indices; in a cubic graph this records exactly the neighbor counts in all ordered cells. Order the resulting cells by that invariant and repeat until stable. If the partition is not discrete, select the first smallest nonsingleton cell and branch over every vertex in it, placing the chosen singleton before the remainder. At a discrete leaf, encode the whole graph in the resulting vertex order. Retain the lexicographically smallest encoding and count the leaves attaining it.

k	Connected classes	$ C_k $	$ S_k $	Orbit mass = D_k
3	1	1	1	1
4	1	1	1	24
5	2	2	2	2,040
6	5	3	4	297,000
7	13	7	8	68,909,400
8	38	15	18	24,038,380,800
9	149	27	32	12,022,907,500,800
10	703	51	61	8,301,377,508,480,000
11	4,132	87	109	7,672,724,783,982,720,000
12	29,579	148	189	9,253,919,577,117,156,480,000

Table 1: Exact connected censuses and spectrum sizes. Every orbit mass equals the connected count obtained without the graph census.

Refinement and cell selection commute with graph isomorphisms. Visiting every choice therefore makes the minimum encoding isomorphism invariant, and equal encodings directly give an isomorphism. Distinct branches give distinct vertex orders: their first different individualization fixes different vertices in the same eventual position. Automorphisms act freely on these orders and preserve the search tree. Any two orders with the same encoding differ by exactly one automorphism. Hence the number of leaves attaining the minimum is exactly $|\text{Aut}(G)|$.

This separate computation checked all 34 623 graphs, with no sampling. Every canonical form within each size was distinct, and all automorphism orders agreed. The resulting orbit masses equal the independently calculated D_k at every required size; Table 1 records them. Lemma 1 proves that all ten lists are exhaustive.

4 Matching counts and the final union

For each connected graph, choose a bipartition and form its $k \times k$ adjacency matrix A . A subset recurrence stores $f(M)$, the number of ways to match the first $|M|$ rows to exactly the column set M . Start with $f(\emptyset) = 1$. For each $j \notin M$ such that $A_{|M|+1,j} = 1$, add $f(M)$ to $f(M \cup \{j\})$. Each partial matching has a unique last choice, so at the full set the result is $\text{per}(A)$.

A separate implementation uses the inclusion–exclusion identity

$$\text{per}(A) = \sum_{J \subseteq \{1, \dots, k\}} (-1)^{k-|J|} \prod_{i=1}^k \sum_{j \in J} A_{ij}. \quad (9)$$

To verify it, expand each product into row-to-column functions. A function with image I has total coefficient $\sum_{J \supseteq I} (-1)^{k-|J|}$, which is zero unless I is the full set, when it is one. The remaining functions are exactly the permutations. Gray-code traversal updates one column at a time and evaluates the same sum. Both implementations agree on every graph. The separate census verifier also recomputed every matching count by a last-row deletion recurrence.

There are precisely nine unordered partitions of 12 with all parts at least three. Their product supports are as follows:

Partition	Values	Partition	Values
(12)	148	(3, 3, 6)	3
(3, 9)	27	(3, 4, 5)	2
(4, 8)	15	(4, 4, 4)	1
(5, 7)	13	(3, 3, 3, 3)	1
(6, 6)	6		

These supports overlap; their union, not their summed cardinalities, is 189. Taking this union in (2), using the exhaustive connected lists, proves Theorem 1.

For an additional check, let $D_k(a)$ be the connected labeled frequency at permanent a , obtained by summing the orbit weights of matching count a . Let $H_n(p)$ be the full labeled frequency, with $H_0(1) = 1$ and zero at other values. The same distinguished-component bijection gives

$$H_n(p) = \sum_{k=1}^n \binom{n-1}{k-1} \binom{n}{k} \sum_{ab=p} D_k(a) H_{n-k}(b). \quad (10)$$

A separate calculation used unordered component partitions instead. For $(k_1, \dots, k_t)$ with multiplicities m_k , its label-distribution factor is

$$\frac{(n!)^2}{\prod_i (k_i!)^2 \prod_k m_k!}.$$

To see this, partition row labels and column labels separately into their blocks, then bijectively pair blocks of equal size. Multiplying the connected frequency distributions, with permanents multiplied, gives the partition's contribution. The two organizations agree at every frequency. Their masses sum to T_n , and the results agree with all saved published frequencies through order eleven. All 189 order-twelve witnesses were additionally checked for entries, margins, component reconstructions, and permanent by separate Python-integer routines, including row/column deletion. All 170 matrices in the saved comparison list were also checked.

5 Reproducibility and limitations

The supplementary dossier supplies the graph6 lists, per-graph certificates, exact spectra, full order-twelve frequency distribution, and one matrix for every spectrum value. In particular, **research-spectrum.json** contains the spectra, **frequencies-12.json** contains the labeled frequencies, and **spectrum-12-witnesses.json** contains the witnesses. The accompanying README gives replay commands for both implementations. The second verification uses Python and a C++17 compiler without a graph-library dependency.

All counting is exact. For $k \leq 12$, a matching count is at most 3^{12} , and the sum of absolute values of the terms in (9) is at most $2^{12}3^{12} = 2\,176\,782\,336$. The implementations use 64-bit integer types for these quantities. Labeled frequencies, orbit-mass sums, and the coefficient calculation use arbitrary-precision integers. Automorphism orders were checked by integer enumeration. There is no floating-point extrapolation.

This is a finite computational proof, with the usual dependencies on correct program implementation, compiler, and execution. It is not a proof-assistant formalization. The independently implemented verification is a separate computational check, not a claim of external peer review. No value of $|S_n|$ for $n > 12$ is asserted. Bibliographic comparison is restricted to the saved sources described above.

Computational and AI assistance. OpenAI Codex assisted in developing the mathematical reductions, writing the exact enumeration and verification programs, inspecting the saved sources, and preparing this manuscript. A separate model context implemented the additional exhaustive checks described here. The mathematical evidence is the stated arguments together with the exact computations and supplied certificates; model agreement itself is not evidence of correctness.

A The exact spectrum

Write $[u, v]_{\mathbb{Z}} = \{u, u + 1, \dots, v\}$. Then

$$S_{12} = [113, 210]_{\mathbb{Z}} \cup E,$$

where the 91 elements of E are listed below. This notation represents $98 + 91 = 189$ distinct integers.

212	213	214	216	217	218	219
220	221	222	224	225	226	228
230	231	232	233	234	236	237
238	240	241	242	243	244	246
248	250	252	256	258	260	264
266	267	268	270	272	274	276
280	282	284	288	289	294	296
297	298	300	306	312	315	318
324	325	330	333	336	338	340
342	348	351	354	356	360	369
372	378	384	390	396	400	403
405	408	416	420	432	441	468
504	612	648	702	720	729	1296

References

- [1] The OEIS Foundation Inc., *A185178: Number of distinct values of the permanent of an $n \times n$ $(0, 1)$ -matrix with exactly three 1's in each row and each column*, entry contributed by Vladimir Shevelev, revision 42, May 30, 2026. <https://oeis.org/A185178>. Saved September 8, 2026.
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